Learning the regulatory grammar of DNA for gene expression engineering

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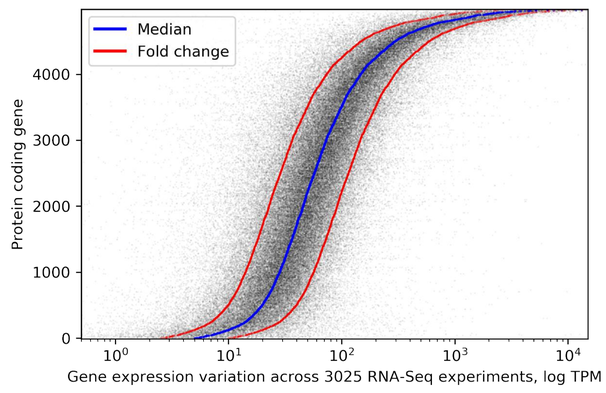
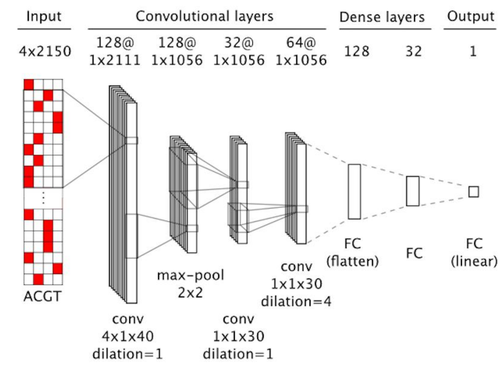
Flash talk @ https://youtu.be/i6foifwt3qU; paper @ https://doi.org/10.1038/s41467-020-19921-4

Expression variation across genes < than whole expression range



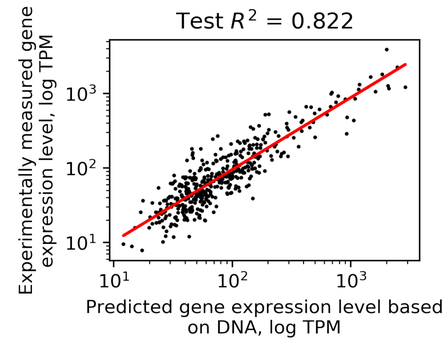
Taking a large fraction of available yeast RNAseq experiments across practically all conditions, and plotting the distributions of RNA expression levels (TPM, x-axis) across sorted genes (y-axis) shows that the observed variation of expression per gene is much smaller than the whole dynamic range of expression levels.

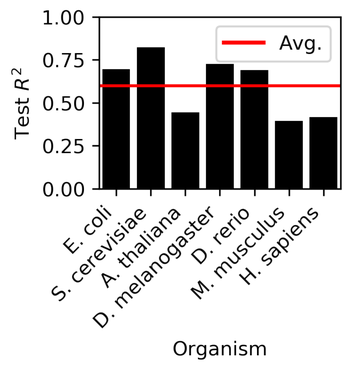
Gene expression levels are encoded in the DNA sequence



Deep neural networks were trained on the gene regulatory structure DNA sequence data (see 2nd box to the left) to predict median gene expression levels.

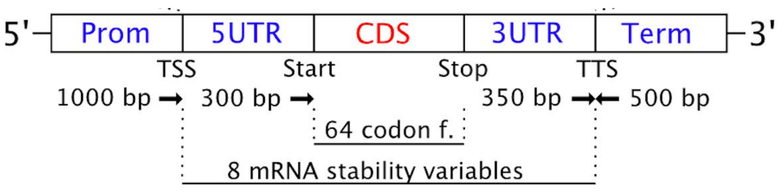
With yeast, **over 82%** of the variation of gene expression levels can be predicted from the DNA sequence alone.





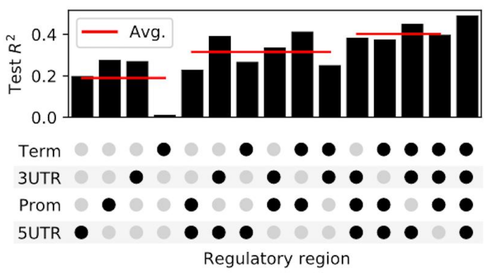
Verifying the phenomenon across multiple model organisms from bacteria to Human shows that on average **60% of gene expression variation is encoded** in the gene regulatory structures of these organisms.

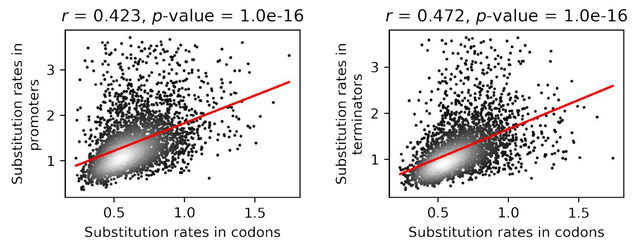
An interacting and co-evolving gene regulatory structure



We defined the **gene regulatory structure** as the coding sequence (CDS) and the non-coding regulatory regions around it spanning the promoter, untranslated regions (UTR) and terminator.

After training the deep neural nets (see upper box to the right), analysis of single contributions showed that **each region** carries some overlapping and orthogonal information and thus **jointly contributes** to gene expression prediction.

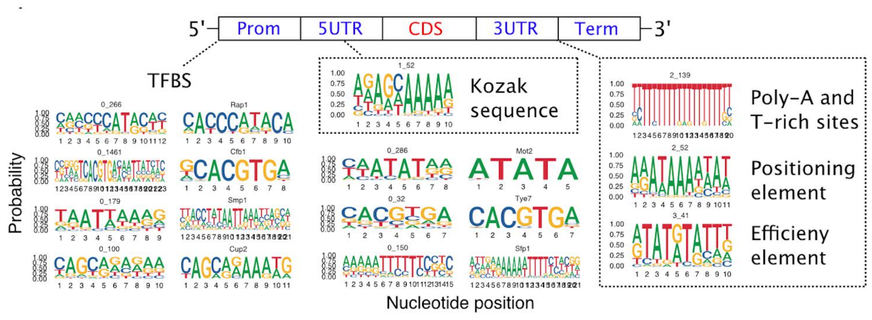




Analysis of mutation rates across ~3000 orthologous genes of 14 related yeast species gives evidence of **co-evolution of regulatory and coding regions**.

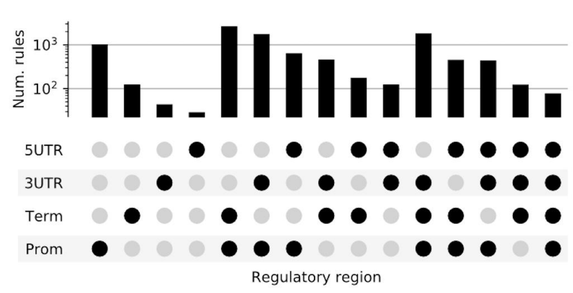
Regulatory motif co-occurrences define gene expression levels

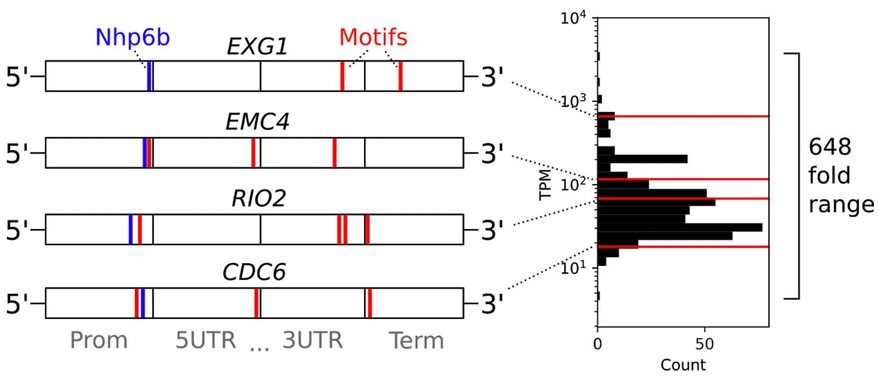
The **regulatory grammar** of gene expression **was extracted** from the neural networks (NN) by using 10bp sliding occlussions of input data and measuring the change in output (defined as relevance score).



Extracted DNA motifs correspond to a variety of **known and expected yeast regulatory grammar** including transcription factor binding sites (TFBS), Kozak sequences and other motifs present in all the different regulatory regions.

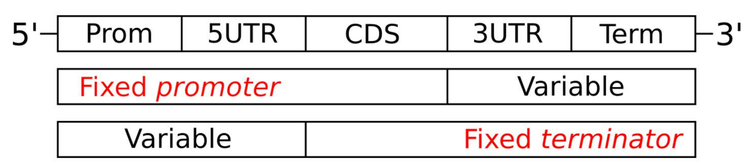
Association rule analysis (used in market basket research), where each gene was treated each as a basket of DNA motifs, uncovered an over 8-fold larger amount of **co-occuring motifs across the different regions** than in any single region.

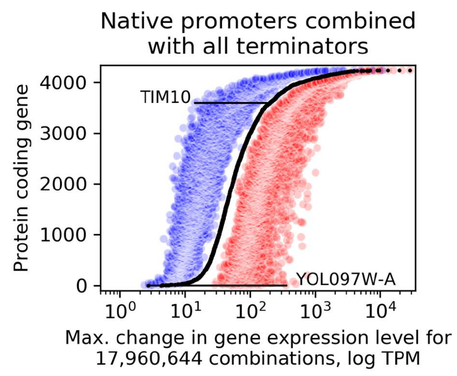




Example of four genes that carry a group of motif co-occurrence rules with a common motif (NHP6B TFBS, blue line) in their promoter regions, and the other 2 - 4 DNA motifs across the remaining regulatory regions (red lines) **repurpose** the gene expression across a 648-fold range of values.

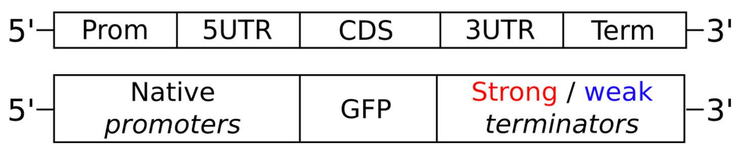
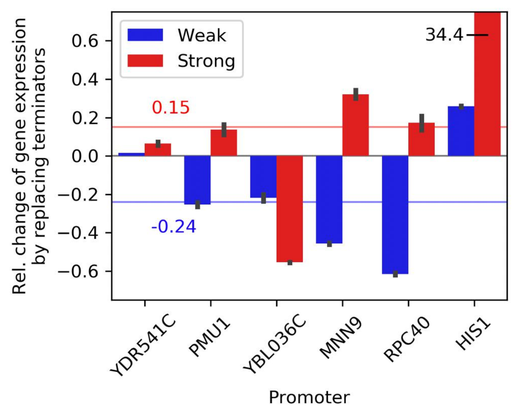
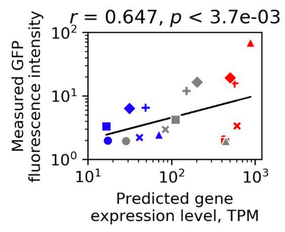
Large degree of regulatory freedom is predicted for each gene





Taking native promoters of genes fixed and swapping the terminators with all available variants shows that the expression levels can be varied **within ~2 orders of magnitude around the native levels**.
Similar results were obtained by conversely varying the promoter regions with fixed terminators.

Learned regulatory grammar guides gene expression engineering



Experimental analysis of constructs with native promoters and terminators swapped with predicted strong/weak variants, shows that expression levels can be **up-/down-regulated by ~20%**.
And it also uncovers constructs with combinations leading to much larger expression changes.